Graph Embedding (GE) - regular assignment

Task 1 (3p.): Provide the quality evaluation of node embeddings obtained by means of Cleora.

- Select the dataset (with labels) and provide it in the CLEORA input scheme; the network dataset may be a graph or hypergraph; choose the task: node classification or link prediction/recommendation,
- produce embeddings and provide their evaluation:
  - Compare performance for different settings of a number of Markov random walk steps, e.g., 3, 5, and 7.

Task 2 (2p.): Visualisation of embeddings. Produce embeddings for a smaller dataset and then reduce the dimension to 2 using UMAP (https://github.com/lmcinnes/umap) or/end t-SNE. Plot them, highlighting the clusters with colors. It is better to use a smaller dataset (e.g., with hundreds of nodes, and shorter embeddings, e.g., vectors of size about 30).

Task 3* (3p). Produce embeddings for the selected 3 embedding algorithms (different than CLEORA), e.g., using the karateclub library or examples from the notebook, and provide their evaluation by means of their performance checking in the selected ML task (classification, prediction, etc) or by means of embedding quality measures (proximity/centrality).

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Useful links:

- Chapter 6 tutorial from "Mining Complex Networks": https://github.com/ftheberge/GraphMiningNotebooks/blob/master/Python_Notebooks_Second_Edition/Chapter_6.ipynb
- Chapter 6 videos on YouTube Channel: https://www.youtube.com/@MiningComplexNetworks
- CLEORA:
  Repository: https://github.com/BaseModelAI/cleora
  Documentation: https://cleora.readthedocs.io/en/latest/
- UMAP:
  https://github.com/lmcinnes/umap

Datasets, which you may use:
GRAPHS:

- Stanford repository
  https://snap.stanford.edu/data/
- Datasets linked at the Dataset Reference section of:
  https://github.com/ftheberge/GraphMiningNotebooks/tree/master
- Austin Benson's repository:
  https://www.cs.cornell.edu/~arb/data/

HYPERGRAPHS:

- Hypernetx basic datasets (GoT, LessMiss, etc.):
  https://github.com/pnnl/HyperNetX/tree/master/hypernetx/utils/toys
- Austin Benson's repository:
  https://www.cs.cornell.edu/~arb/data/

- XGI repository:
  https://xgi.readthedocs.io/en/stable/xgi-data.html